

Unknown is not normal: separating language-model extraction from rule-based decision logic for clinical risk scores

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Abstract

Large language models (LLMs) are increasingly used to compute clinical risk scores from free-text notes. Notes are often incomplete, and treating undocumented findings as normal can silently misclassify patients. We test whether separating three-state extraction (present, absent or unknown, by an LLM) from decision logic (deterministic code computing score bounds over unknown inputs) lets a system ask only questions that can change the decision. On 1,200 synthetic emergency cases across six calculators (HEART, CURB-65, qSOFA, PERC, Wells, Cockcroft-Gault), with a simulated clinician answering questions, we compared this bounds policy with asking for every missing input, a missing-equals-normal schema, and an end-to-end LLM agent (Claude Opus 5.5). With Claude Haiku 4.5 as extractor, the bounds policy matched ask-all accuracy (99.4% vs 99.4%) with half the questions (0.92 vs 1.78 per case) and no irrelevant ones. Treating missing as normal dropped accuracy to 91.2% and under-triaged 8.5% of patients (95% CI 7.1–10.2), and under-triage persisted under messy notes and a noisy clinician. The agent was equally accurate under ideal conditions (99.6%) but 9.5% of its questions were irrelevant; with a noisy clinician it was less accurate than the bounds policy (83.5% vs 87.0%, $p < 0.001$) and committed prematurely in 2.7% of cases (bounds: 0%). A 9B local model as extractor reached oracle-level accuracy (99.8%). In 584 real case reports from MedCalc-Bench, only 52% contained enough information to determine the category (HEART 13%). Routing decisions through code that reasons explicitly about unknowns avoids premature commitment and irrelevant questions, halves the questions asked, and works with small local models.

1. Introduction

Clinical scores such as HEART (chest pain), CURB-65 (pneumonia), qSOFA (suspected sepsis), PERC and Wells (pulmonary embolism) and Cockcroft-Gault (renal dosing) are defined over a few inputs with explicit thresholds. What matters clinically is usually the *decision category*, not the exact score: a HEART score of 0–3 is low risk, 4–6 moderate, 7 or more high. LLMs are now used to compute these scores from notes ([Khandekar et al. 2024](#); [Wang et al. 2025](#); [Zhu et al. 2026](#)), and the standard benchmark, MedCalc-Bench, treats an input the note does not mention as absent or normal ([Khandekar et al. 2024](#)).

That convention is not neutral. “No mention of hemoptysis” is not “no hemoptysis”, and a score computed as if it were can place a patient in a lower-risk category. The safe behaviour is to recognise when the documented facts do not determine the decision and then ask only for facts that could change it. LLMs handle this poorly in both directions, committing when the category is undetermined and abstaining when it is determined ([Watanabe et al. 2026](#)), and interactive benchmarks show that models gathering information ask too little or too much ([Li et al. 2024](#); [Schmidgall et al. 2024](#); [Johri et al. 2025](#)).

We test a simple division of labour (Figure 1). A language model reads the note into three-valued facts (present with a value, explicitly absent, or unknown), each with an exact evidence quote and a confidence. Deterministic code computes the range of scores still possible given the unknowns (the score’s *bounds*), decides whether the category is determined, and if not asks only about unknowns that could change it. We compare this with asking for every missing input, with the same pipeline treating missing as normal, and with an end-to-end LLM agent. We pre-specified four hypotheses: (H1) the bounds policy asks far fewer questions than ask-all at equal accuracy; (H2) an end-to-end agent both asks questions that cannot change the category and answers before the category is determined; (H3)

three-valued extraction reduces silent missing-as-absent errors compared with a binary schema; (H4) confirming low-confidence, decision-critical values catches extraction errors at a small question cost.

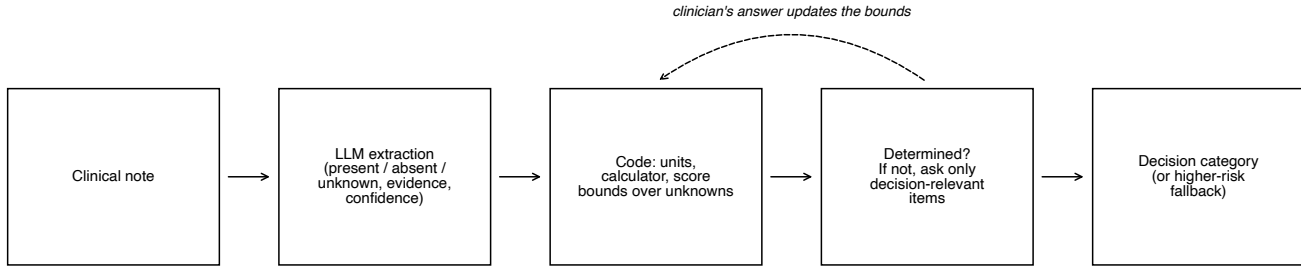


Figure 1: Pipeline. A language model reads the note into three-valued facts with evidence and confidence. Code normalises units, computes the score’s bounds over the unknowns and, if the decision category is not determined, asks the clinician only about inputs that could change it; each answer updates the bounds.

2. Methods

Calculators and bounds. The six calculators were implemented from their primary sources (Six et al. 2008; Lim et al. 2003; Seymour et al. 2016; Singer et al. 2016; Kline et al. 2004, 2008; Wells et al. 2000; van Belle et al. 2006; Cockcroft and Gault 1976), with decision categories HEART 0–3 / 4–6 / ≥ 7 , CURB-65 0–1 / 2 / ≥ 3 , qSOFA ≥ 2 positive, PERC negative if all criteria are met, Wells PE likely if > 4 , and Cockcroft-Gault < 30 / 30 – < 60 / ≥ 60 mL/min. Where a source was ambiguous (for example, HEART troponin bands or the BUN equivalent of CURB-65 urea) we implemented a documented reading; the author, an emergency physician, reviewed all readings (Supplementary S1). Units are converted by code, never by the model. Inputs are boolean, ordinal or numeric; numeric inputs are split into regions at the calculator’s cut-points, and achievable scores are enumerated exactly over the regions (Cockcroft-Gault, which is monotone, is evaluated at interval corners). The category is *determined* when only one category is possible, and an unknown input is *decision-relevant* if its value changes the category for some completion of the other unknowns. For example, with 2 documented HEART points and troponin (0–2 points) unknown, the possible scores 2–4 span low and moderate risk, so troponin is asked; with 0 documented points they do not, and nothing is asked. Property-based tests (Maclver et al. 2019) checked that every completion falls within the bounds and that inputs labelled irrelevant never change the category.

Synthetic cohort and notes. For each calculator we sampled 200 patients from plausible emergency-department priors (Supplementary S2). Each input was left undocumented with probability 0.3, and cases were resampled so that half had a category the note did not determine; results can be reweighted to the natural share (Supplementary S3). Cases carried traps: negations, values in other units, values from a previous visit, superseded readings, and comorbidities implied only by a medication. Claude Sonnet 5 wrote each note from a fact sheet, stating documented values exactly and omitting undocumented inputs, including indirect cues. Every note passed rule checks and an independent LLM judge (Claude Opus 5.5) comparing each input with the ground truth (22 of 1,200 needed one retry). A “messy” end-of-shift version (fragments, abbreviations, typos, copied-forward text) was rendered for 600 cases; 559 passed validation and form the paired set for harder conditions.

Extraction. For every input the extractor returned a status, a value, a unit and the shortest exact supporting quote, plus a confidence. Code rejected any claim whose quote was not an exact substring of the note (the input then became unknown), converted units, and rejected implausible values. Extractors were an oracle returning the documented facts (a perfect-extraction reference), Claude Haiku 4.5 (through headless Claude Code with tools disabled and schema-constrained output; self-reported confidence), and Qwen3.5-9B (Qwen Team 2026) run locally through Ollama (Ollama 2026) (confidence from token log-probabilities). Confidence was calibrated by temperature scaling (Guo et al. 2017) and isotonic regression (Zadrozny and Elkan 2002) on a 30% development split.

Policies. S1 asks about every input left unknown. S3 (bounds) asks only decision-relevant unknowns, one at a time, recomputing the bounds after each answer. S4 adds value-of-information ordering and asks the clinician to confirm

decision-critical values with calibrated confidence below 0.9. S3-bin is S3 with missing inputs treated as normal, the MedCalc-Bench convention. S2 is an end-to-end agent: Claude Opus 5.5 receives the note and the input list and, turn by turn, asks the clinician, runs the calculator, or answers (or declares the case undeterminable). When no question can settle the category, a policy abstains; for secondary analyses an abstention becomes the highest-risk category still possible (for the agent, the calculator’s highest-risk category).

Simulated clinician. A deterministic simulator answered from the hidden truth; troponin was not yet available in 10% of cases. The noisy clinician could not answer 10% of questions, answered wrongly 5% of the time (a flipped yes/no, a graded input one level off, or a number off by 10–25%), and gave 25% of numeric answers as a $\pm 10\%$ range. The noise is fixed per case and input, so all policies face the same clinician.

Outcomes and statistics. Primary outcomes were decision-category accuracy, with abstention counted as incorrect, and questions per case. Secondary outcomes were under-triage (a lower-risk category than the truth) and over-triage; premature commitment (answering while the category was undetermined); irrelevant questions (whose answer could not change the category); silent missing-as-absent claims; extraction accuracy; and calibration. A comorbidity implied only by a medication is not determined by the note, so “present” and “unknown” were both accepted there. Proportions have Wilson intervals (Wilson 1927) and mean questions bootstrap intervals. Paired comparisons used exact McNemar (McNemar 1947) and Wilcoxon signed-rank tests, Holm-adjusted (Holm 1979) across four pre-specified comparisons per condition.

Real notes. We used MedCalc-Bench Verified (Khandekar et al. 2024) and its label audits (Ye et al. 2025; Krohn-Grimberghe 2026) for the five overlapping calculators: 100 test notes and a sample of 584 training notes (published, de-identified case reports; CC-BY-SA 4.0). We report whether our code reproduces the benchmark labels from its annotated inputs, how often the note determines the category, whether the truth stays within our bounds, and accuracy under the missing-equals-normal convention.

Implementation. All runs cost \$0 in API fees (Claude through a subscription; Qwen on a 16 GB laptop). Each run is one configuration file with a fixed seed, and every model call is cached, so all tables and figures regenerate without new model calls. Code: <https://github.com/nicoveraz/calc-bounds> (MIT licence).

3. Results

Ideal conditions (Table 1). H1 was supported. The bounds policy matched ask-all accuracy with every extractor while asking 44–48% fewer questions (Haiku: 99.4% for both, 0.92 vs 1.78 questions; $p < 0.001$). About half of ask-all’s questions could not change the decision; the bounds policy asked none. The saving held at 10%, 30% and 50% missingness (34–49% fewer questions; Supplementary S3). H3 was supported: treating missing as normal (S3-bin) reduced accuracy to 91.2–91.8% ($p < 0.001$), answered before the category was determined in 39–49% of cases, and under-triaged 8.2–8.8% of patients, against 0.0–0.1% for the bounds policy. Haiku itself occasionally turned an undocumented input into “absent” (0.022 per case), which explains most of its small accuracy loss. Qwen missed 17% of documented negatives but labelled them unknown, so they were asked rather than assumed. H2 was partly supported: the agent was as accurate (99.6%) but asked more questions (0.99 per case, $p < 0.001$), 9.5% of them irrelevant, and answered prematurely in 0.2% of cases. H4 was partly supported: confirmation raised Haiku accuracy from 99.4% to 99.7% with fewer questions overall (0.90 vs 0.92), but the gain was not significant because Haiku made few errors to catch. Calibration reduced expected calibration error to 0.002 for both extractors (Supplementary S5).

Table 1. Clean notes, ideal clinician ($n = 1,200$). S1 ask-all; S2 end-to-end agent (reads the note, so identical across extractors); S3 bounds; S4 bounds with value-of-information ordering and confirmation; S3-bin missing = normal. Accuracy counts abstention as incorrect; under-triage uses the higher-risk fallback.

Extraction		Accuracy, % System (95% CI)	Questions/case (95% CI)	Irrelevant, %	Premature, %	Under-triage, % (95% CI)
Oracle	S1	99.8 (99.4–100.0)	1.74 (1.67–1.82)	47.3	0.0	0.0 (0.0–0.3)
Oracle	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)

Extraction	System	Accuracy, % (95% CI)	Questions/case (95% CI)	Irrelevant, %	Premature, %	Under-triage, % (95% CI)
Oracle	S3	99.8 (99.4–100.0)	0.92 (0.85–0.99)	0.0	0.0	0.0 (0.0–0.3)
Oracle	S4	99.8 (99.4–100.0)	0.87 (0.81–0.93)	0.0	0.0	0.0 (0.0–0.3)
Oracle	S3-bin	91.8 (90.1–93.2)	0.21 (0.19–0.24)	0.0	39.2	8.2 (6.8–9.9)
Haiku 4.5	S1	99.4 (98.8–99.7)	1.78 (1.71–1.86)	48.1	0.0	0.1 (0.0–0.5)
Haiku 4.5	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)
Haiku 4.5	S3	99.4 (98.8–99.7)	0.92 (0.86–0.99)	0.0	0.0	0.1 (0.0–0.5)
Haiku 4.5	S4	99.7 (99.1–99.9)	0.90 (0.83–0.96)	0.0	0.0	0.1 (0.0–0.5)
Haiku 4.5	S3-bin	91.2 (89.4–92.6)	0.21 (0.18–0.23)	0.0	40.0	8.5 (7.1–10.2)
Qwen3.5-9B (local)	S1	99.8 (99.4–100.0)	2.21 (2.12–2.29)	44.3	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)
Qwen3.5-9B (local)	S3	99.8 (99.4–100.0)	1.23 (1.15–1.31)	0.0	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S4	99.8 (99.4–100.0)	1.20 (1.13–1.27)	0.0	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S3-bin	91.2 (89.5–92.7)	0.26 (0.23–0.29)	0.0	49.2	8.8 (7.3–10.5)

Harder conditions (Table 2, Figures 2–3). A noisy clinician cost every policy that asks 12–16 accuracy points, mostly through abstention: a “don’t know” to the one decisive question leaves the category open. When they did answer, S1, S3 and S4 were 96.8–97.8% correct. The agent became less accurate than the bounds policy (83.5% vs 87.0%, $p < 0.001$) and committed prematurely in 2.7% of cases; the code policies never did. S3-bin’s overall accuracy matched the bounds policy under noise because it rarely asks, but when it answered it was right only 89–91% of the time, and it under-triaged 8.4–10.2% of patients in every condition (the bounds policy: 0.0–0.5%). With the higher-risk fallback, the bounds policies over-triaged when information was genuinely unavailable (9.6–12.9% under noise), the safe direction. On messy notes, Haiku extraction lost a little accuracy (98.6%; S4 recovered 99.1%), Qwen none (99.5%), and the agent reading the notes directly was more accurate than S3 with Haiku (99.8% vs 98.6%, $p = 0.047$).

Table 2. Harder conditions (notes / clinician), Haiku extraction, 559 paired cases; policies as in Table 1. Full grid with Qwen3.5-9B in Supplementary S6.

Condition	System	Accuracy, % (95% CI)	Correct if answered, %	Under-triage, % (95% CI)	Over- triage, %	Premature, %	Questions/case
clean / ideal	S1	99.1 (97.9–99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	1.71 (1.60–1.81)
clean / ideal	S2	99.6 (98.7–99.9)	99.8	0.2 (0.0–1.0)	0.2	0.2	0.92 (0.82–1.01)
clean / ideal	S3	99.1 (97.9–99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	0.85 (0.75–0.94)
clean / ideal	S4	99.6 (98.7–99.9)	99.8	0.2 (0.0–1.0)	0.2	0.0	0.82 (0.73–0.91)
clean / ideal	S3-bin	91.4 (88.8–93.5)	91.4	8.2 (6.2–10.8)	0.4	37.0	0.20 (0.16–0.24)

Condition	System	Accuracy, % (95% CI)	Correct if answered, %	Under-triage, % (95% CI)	Over- triage, %	Premature, %	Questions/case
clean / noisy	S1	88.2 (85.3–90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	1.71 (1.60–1.81)
clean / noisy	S2	84.4 (81.2–87.2)	97.7	0.5 (0.2–1.6)	13.1	1.8	0.95 (0.84–1.05)
clean / noisy	S3	88.2 (85.3–90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	0.88 (0.77–0.97)
clean / noisy	S4	88.7 (85.8–91.1)	98.0	0.5 (0.2–1.6)	8.2	0.0	0.86 (0.76–0.96)
clean / noisy	S3- bin	88.7 (85.8–91.1)	91.2	8.2 (6.2–10.8)	2.5	36.0	0.20 (0.16–0.25)
messy / ideal	S1	98.6 (97.2–99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.92 (1.80–2.04)
messy / ideal	S2	99.8 (99.0–100.0)	100.0	0.0 (0.0–0.7)	0.2	0.7	0.92 (0.82–1.01)
messy / ideal	S3	98.6 (97.2–99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.04 (0.94–1.15)
messy / ideal	S4	99.1 (97.9–99.6)	99.3	0.5 (0.2–1.6)	0.4	0.0	1.04 (0.94–1.15)
messy / ideal	S3- bin	90.7 (88.0–92.8)	90.7	8.8 (6.7–11.4)	0.5	39.4	0.30 (0.23–0.36)
messy / noisy	S1	85.5 (82.3–88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.92 (1.80–2.04)
messy / noisy	S2	85.0 (81.8–87.7)	97.3	1.1 (0.5–2.3)	12.0	3.4	0.97 (0.87–1.08)
messy / noisy	S3	85.5 (82.3–88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.08 (0.97–1.20)
messy / noisy	S4	85.9 (82.7–88.5)	97.4	0.5 (0.2–1.6)	10.7	0.0	1.11 (1.00–1.23)
messy / noisy	S3- bin	86.8 (83.7–89.3)	90.3	8.6 (6.5–11.2)	3.8	38.3	0.31 (0.25–0.38)

Real notes (Figure 4). Our code reproduced MedCalc-Bench’s labels from its annotated inputs for all HEART, CURB-65, PERC and Wells notes; for Cockcroft-Gault only 51%, because the benchmark chooses actual, ideal or adjusted weight by body-mass index. Only 52% (95% CI 48–56) of the 584 training case reports had a category determined by the documented facts (Qwen: 47%), ranging from 90% for PERC to 13% for HEART. Under the missing-equals-normal convention the category was correct for 69% of notes (HEART 22%). The truth stayed within our bounds for 93–95% of notes; the misses were extraction errors, mainly in HEART inputs. The test split gave similar results (Supplementary S8).

4. Discussion

The most consistent finding is a safety one: treating undocumented findings as normal under-triaged about one patient in twelve in every condition, including clean notes and a perfect clinician, and gave the right category for only two-thirds of real case reports. Headline accuracy hides this, because a system that rarely asks rarely meets an unanswerable question. Letting the model only read, and code decide, removed premature commitment and irrelevant questions and halved the questions asked. A frontier agent was a strong baseline, as accurate under ideal conditions and more accurate than a small extractor on messy notes, so we do not claim the modular design is more accurate than a capable agent. Its advantages are that it never answers early, never asks an irrelevant question, has the lowest under-triage, is deterministic and auditable, and needs only a model that can read: a 9B local model matched perfect

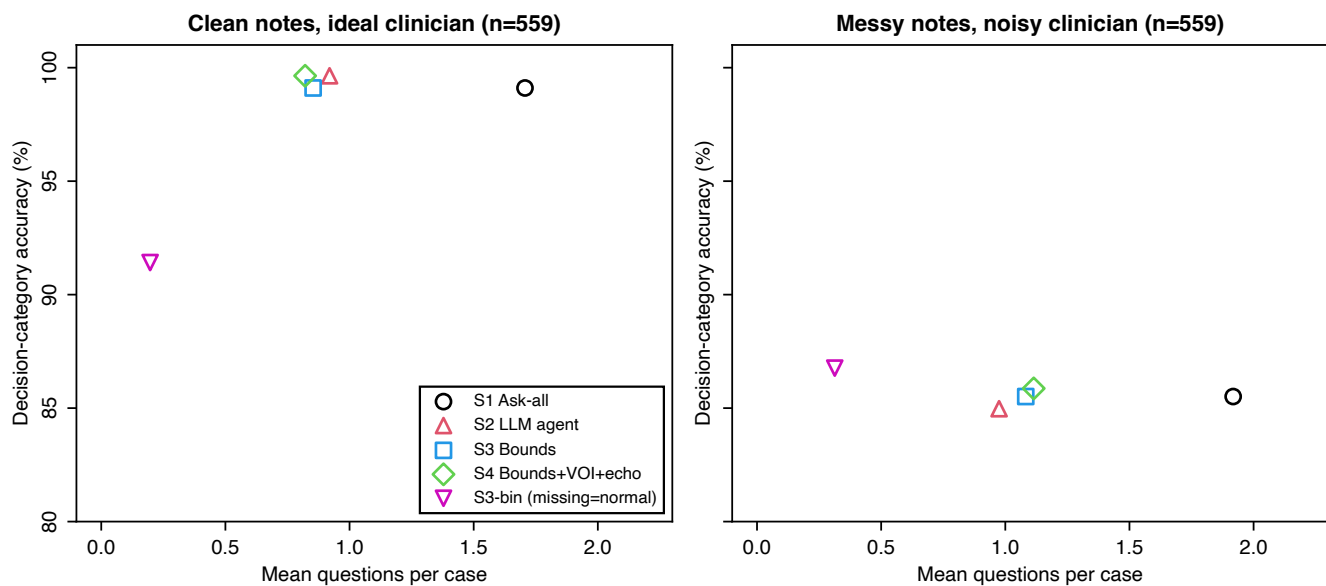


Figure 2: Accuracy versus mean questions per case (Haiku extraction, 559 paired cases). Left: clean notes, ideal clinician. Right: messy notes, noisy clinician.

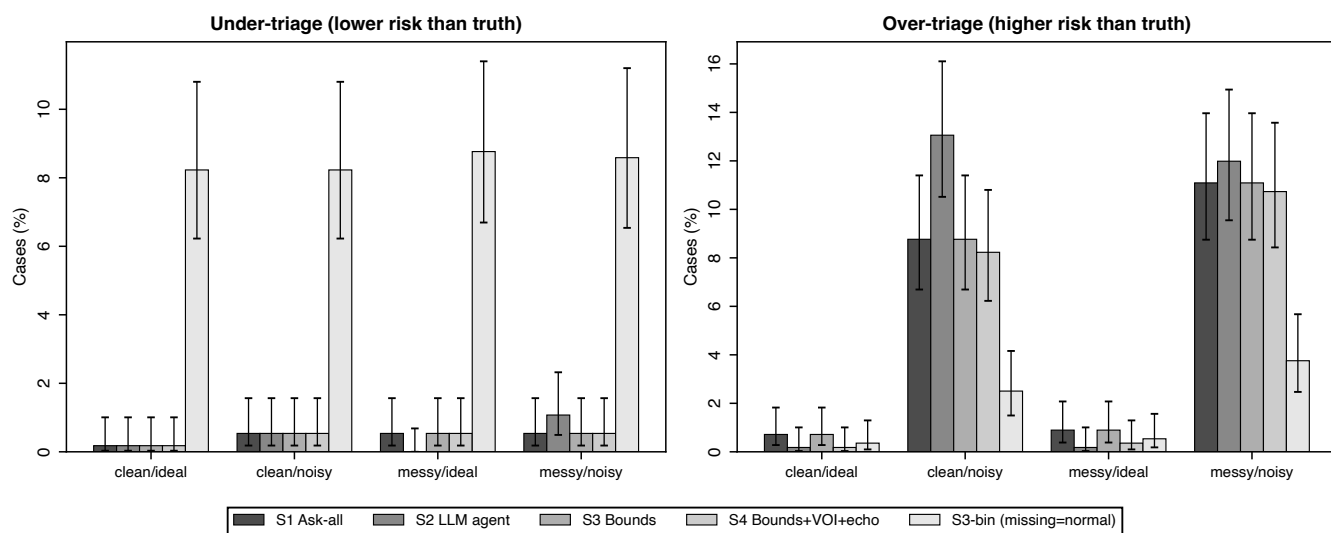


Figure 3: Under-triage and over-triage with a higher-risk fallback for undetermined cases (Haiku extraction, 559 paired cases). Error bars: Wilson 95% confidence intervals.

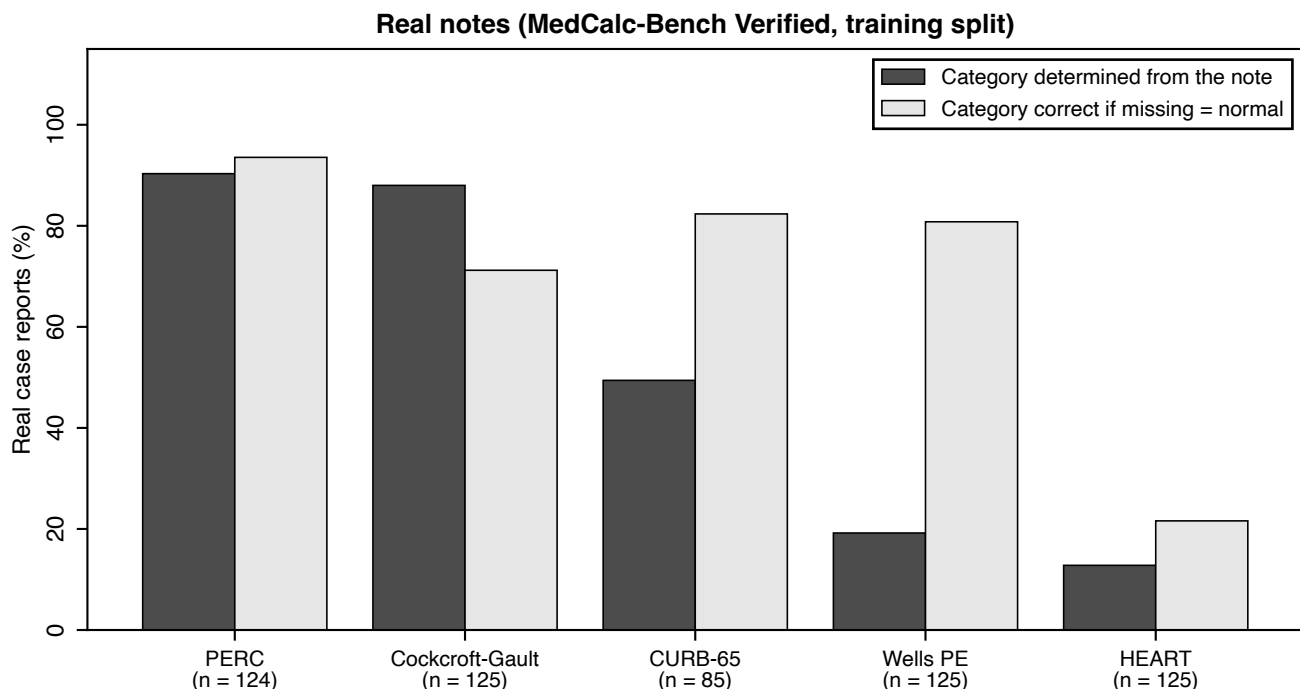


Figure 4: Real case reports (MedCalc-Bench Verified, training split; Haiku extraction): share whose category is determined by the note, and share whose category is correct when missing inputs are treated as normal.

extraction, whereas the agent needs a frontier model at every turn.

ClinDet-Bench (Watanabe et al. 2026) poses the same determinacy question over score ranges; we add question asking and a comparison with an agent that can ask. MediQ, AgentClinic and CRAFT-MD evaluate information seeking in diagnostic dialogue rather than calculators (Li et al. 2024; Schmidgall et al. 2024; Johri et al. 2025). Our real-note analysis suggests that MedCalc-Bench’s convention (Khandekar et al. 2024) overstates how often a score can safely be computed from the note alone. The design relates to active feature acquisition (Saar-Tsechansky et al. 2009), selective prediction (Geifman and El-Yaniv 2017) and partial evaluation (Jones et al. 1993), and executable guideline standards such as Clinical Quality Language (HL7 International 2026a) and FHIR Clinical Practice Guidelines (HL7 International 2026b) are a natural home for the determinacy check.

Limitations. The notes are synthetic, cleaner and more complete than real ones, and were written by a model from the same family as the Claude extractor; a small exploratory comparison with Qwen-written notes suggested a same-family advantage of 3.2 points (95% CI 0.4–6.4). Absolute accuracies are therefore optimistic; question counts and safety differences are the more transferable results. The clinician is simulated with a simple noise model. Patient characteristics were sampled independently, which produced 12 implausible HEART cases; excluding them changed nothing (Supplementary S10). Claude models were used through headless Claude Code, and their confidence is self-reported. The real-note analysis could not include question asking, and its labels have known errors (Ye et al. 2025; Krohn-Grimberghe 2026). Calculator readings and priors were reviewed by one physician. Accuracy is near ceiling under ideal conditions, so several accuracy comparisons are underpowered.

Implications. Keep “unknown” distinct from “absent” end to end, let code decide whether the documented facts determine the decision, ask only what can change it, and fall back to the higher-risk category when information cannot be obtained. Benchmarks that fill missing inputs with normal values should also report how often the note actually determines the category.

Declarations

Data and code. Code: <https://github.com/nicoveraz/calc-bounds> (MIT licence). The synthetic cohort is regenerated from the configuration and seed; notes, extractions and the model-response cache are not released, and re-running regenerates them (model outputs may differ slightly). MedCalc-Bench Verified is available from its authors. **Funding:** none. **Competing interests:** none. **Use of AI:** LLMs generated the synthetic notes and served as extractors, judge and agent; code and analysis were developed with AI assistance (Claude Code). **Ethics:** synthetic data and published de-identified case reports only; no ethics review was required.

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